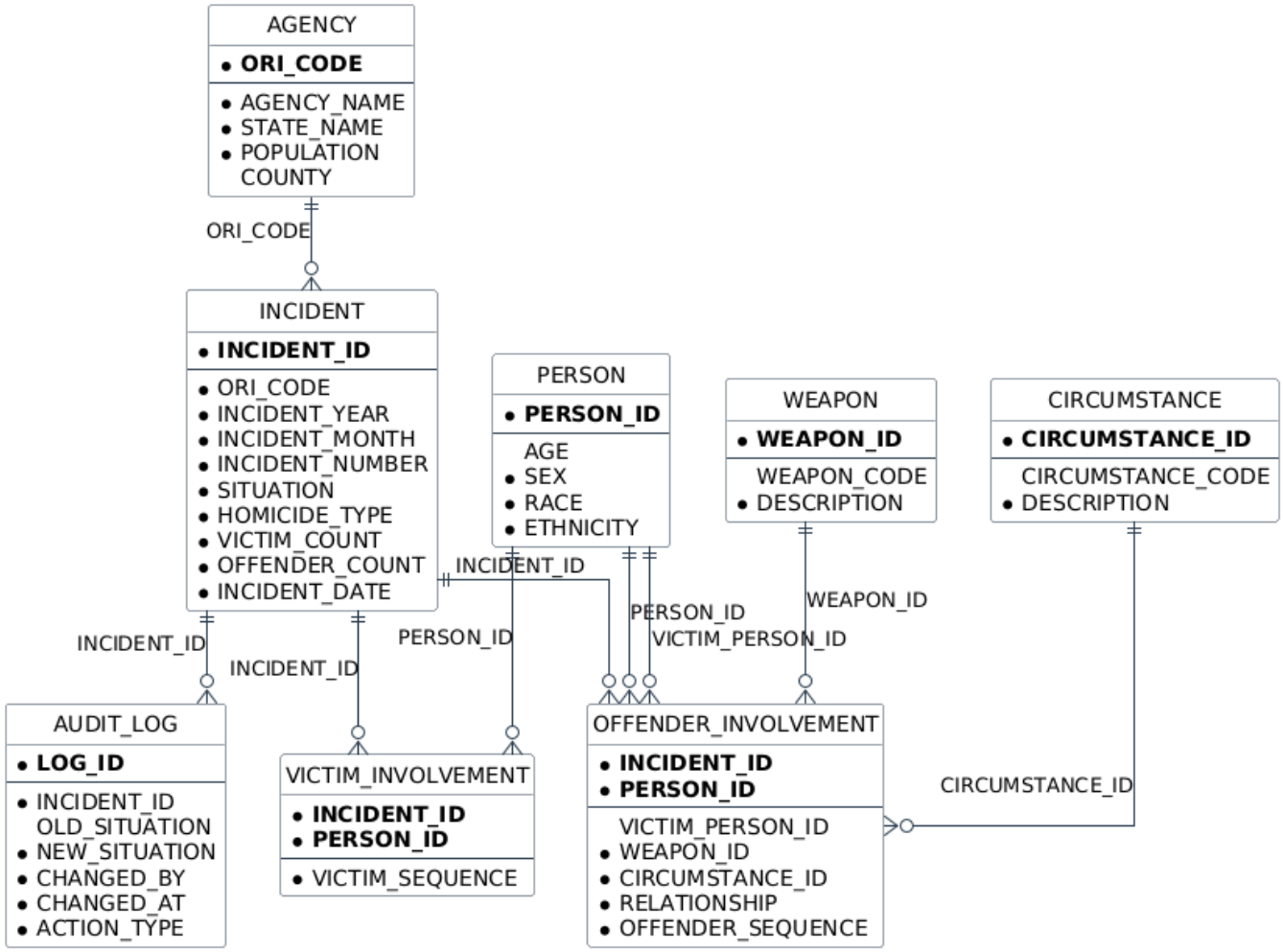
Meilenstein 4.2: SQL & Trigger – US Homicides (SHR)

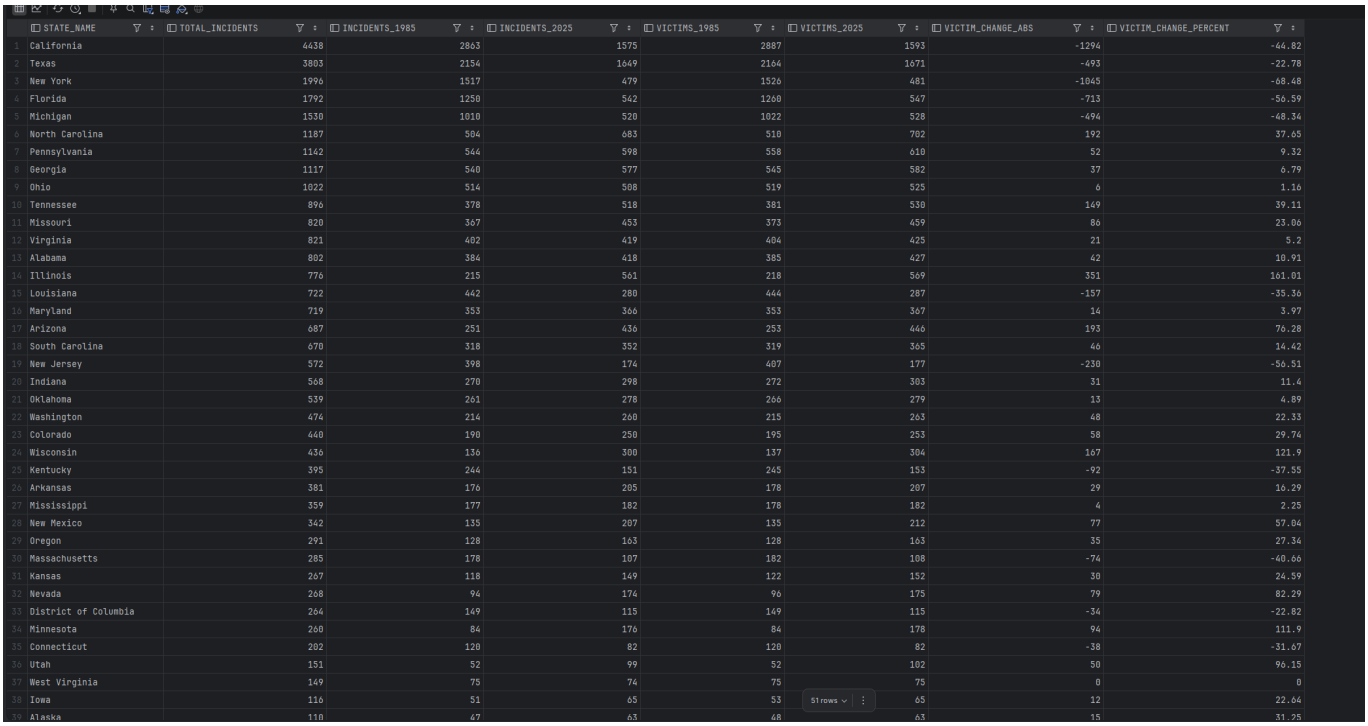
1. Aktualisiertes ERD (Kapitel 2.3)

- **Feedback-Umsetzung:**
  - `victim_person_id` in `OFFENDER_INVOLVEMENT` ergänzt (Bezug zu konkretem Opfer).
  - Definition ungelöst: `situation LIKE '%Unknown Offender%'` bzw. `offender_count = 0`.
  - Eigene Tabelle `AUDIT_LOG` für den After-Update-Trigger ergänzt.



2. SQL-Abfragen (Kapitel 2.5)

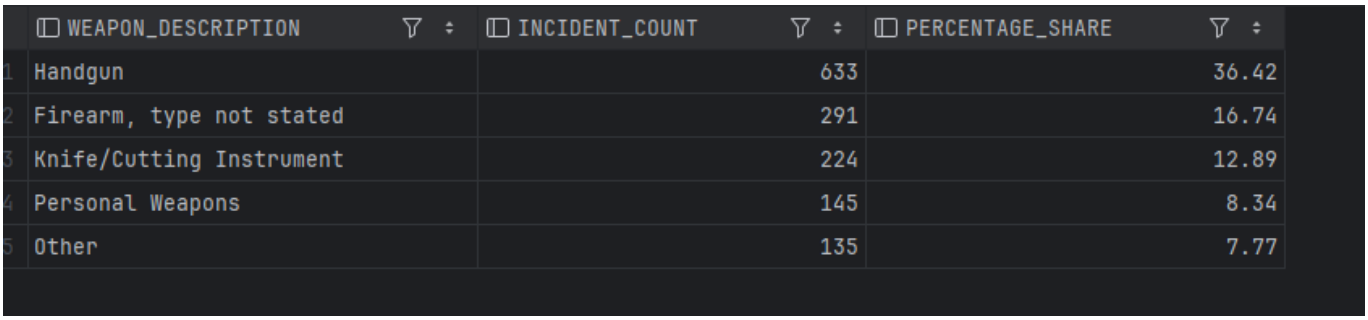
2.1 Abfrage 1: Vergleich Tötungsdelikte und Gesamtopferzahlen (1985 vs. 2025)



	STATE_NAME	TOTAL_INCIDENTS	INCIDENTS_1985	INCIDENTS_2025	VICTIMS_1985	VICTIMS_2025	VICTIM_CHANGE_ABS	VICTIM_CHANGE_PERCENT
1	California	4438	2863	1575	2887	1593	-1294	-44.82
2	Texas	3803	2154	1649	2164	1671	-493	-22.78
3	New York	1996	1517	479	1526	481	-1045	-68.48
4	Florida	1792	1250	542	1260	547	-713	-56.59
5	Michigan	1530	1010	520	1022	528	-494	-48.34
6	North Carolina	1187	504	683	510	702	192	37.65
7	Pennsylvania	1142	544	598	558	610	52	9.32
8	Georgia	1117	540	577	545	582	37	6.79
9	Ohio	1022	514	508	519	525	6	1.16
10	Tennessee	896	378	518	381	530	149	39.11
11	Missouri	820	367	453	373	459	86	23.06
12	Virginia	821	402	419	404	425	21	5.2
13	Alabama	802	384	418	385	427	42	10.91
14	Illinois	776	215	561	218	569	351	161.01
15	Louisiana	722	442	280	444	287	-157	-35.36
16	Maryland	719	353	366	353	367	14	3.97
17	Arizona	687	251	436	253	446	193	76.28
18	South Carolina	670	318	352	319	365	46	14.42
19	New Jersey	572	398	174	407	177	-230	-56.51
20	Indiana	568	270	298	272	303	31	11.4
21	Oklahoma	539	261	278	266	279	13	4.89
22	Washington	474	214	260	215	263	48	22.33
23	Colorado	440	190	250	195	253	58	29.74
24	Wisconsin	436	136	300	137	304	167	121.9
25	Kentucky	395	244	151	245	153	-92	-37.55
26	Arkansas	381	176	205	178	207	29	16.29
27	Mississippi	359	177	182	178	182	4	2.25
28	New Mexico	342	135	207	135	212	77	57.04
29	Oregon	291	128	163	128	163	35	27.34
30	Massachusetts	285	178	107	182	108	-74	-40.66
31	Kansas	267	118	149	122	152	30	24.59
32	Nevada	268	94	174	96	175	79	82.29
33	District of Columbia	264	149	115	149	115	-34	-22.82
34	Minnesota	260	84	176	84	178	94	111.9
35	Connecticut	202	120	82	120	82	-38	-31.67
36	Utah	151	52	99	52	102	50	96.15
37	West Virginia	149	75	74	75	75	0	0
38	Iowa	116	51	65	53	65	12	22.64
39	Alaska	110	47	63	48	63	15	31.25

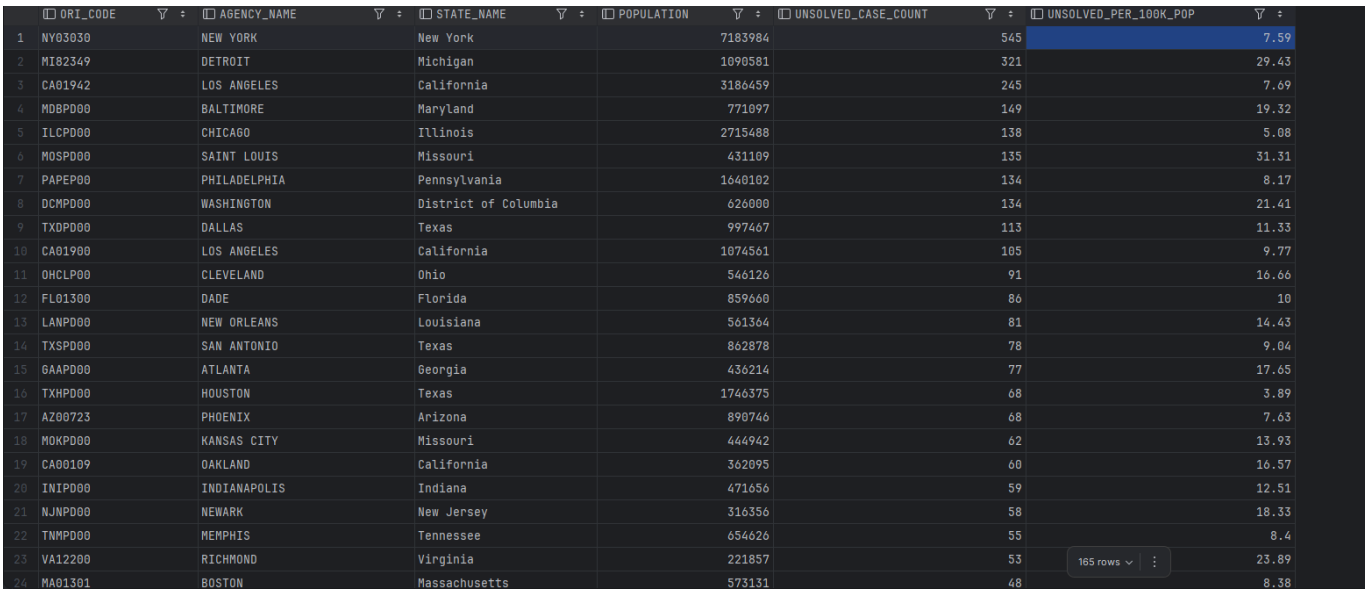
51 rows

2.2 Abfrage 2: Top 5 Tatwaffen bei Beziehungstaten in Staaten mit hoher Fallzahl



	WEAPON_DESCRIPTION	INCIDENT_COUNT	PERCENTAGE_SHARE
1	Handgun	633	36.42
2	Firearm, type not stated	291	16.74
3	Knife/Cutting Instrument	224	12.89
4	Personal Weapons	145	8.34
5	Other	135	7.77

2.3 Abfrage 3: Polizeibehörden mit überdurchschnittlich vielen ungelösten Fällen



	ORI_CODE	AGENCY_NAME	STATE_NAME	POPULATION	UNSOLVED_CASE_COUNT	UNSOLVED_PER_100K_POP
1	NY03030	NEW YORK	New York	7183984	545	7.59
2	MI82349	DETROIT	Michigan	1090581	321	29.43
3	CA01942	LOS ANGELES	California	3186459	245	7.69
4	MDBPD00	BALTIMORE	Maryland	771097	149	19.32
5	ILCPD00	CHICAGO	Illinois	2715488	138	5.08
6	MOSPD00	SAINT LOUIS	Missouri	431109	135	31.31
7	PAPEP00	PHILADELPHIA	Pennsylvania	1640102	134	8.17
8	DCMPD00	WASHINGTON	District of Columbia	626000	134	21.41
9	TXDPD00	DALLAS	Texas	997467	113	11.33
10	CA01900	LOS ANGELES	California	1074561	105	9.77
11	OHCLP00	CLEVELAND	Ohio	546126	91	16.66
12	FL01300	DADE	Florida	859660	86	10
13	LANPD00	NEW ORLEANS	Louisiana	561364	81	14.43
14	TXSPD00	SAN ANTONIO	Texas	862878	78	9.04
15	GAAPD00	ATLANTA	Georgia	436214	77	17.65
16	TXHPD00	HOUSTON	Texas	1746375	68	3.89
17	AZ00723	PHOENIX	Arizona	890746	68	7.63
18	MOKPD00	KANSAS CITY	Missouri	444942	62	13.93
19	CA00109	OAKLAND	California	362095	60	16.57
20	INIPD00	INDIANAPOLIS	Indiana	471656	59	12.51
21	NJNPD00	NEWARK	New Jersey	316356	58	18.33
22	TNMPD00	MEMPHIS	Tennessee	654626	55	8.4
23	VA12200	RICHMOND	Virginia	221857	53	23.89
24	MA01301	BOSTON	Massachusetts	573131	48	8.38

165 rows

3. Data Dictionary Abfragen (Kapitel 2.5)

3.1 Alle Tabellen und Spaltenanzahl

	TABLE_NAME	COLUMN_COUNT	TABLESPACE_NAME	STATUS
1	AGENCY	5	USERS	VALID
2	AUDIT_LOG	7	USERS	VALID
3	CIRCUMSTANCE	3	USERS	VALID
4	INCIDENT	10	USERS	VALID
5	OFFENDER_INVOLVEMENT	7	USERS	VALID
6	PERSON	5	USERS	VALID
7	VICTIM_INVOLVEMENT	3	USERS	VALID
8	WEAPON	3	USERS	VALID

3.2 Spalten, Datentypen und Constraints für Tabelle INCIDENT

Spalten und Datentypen:

	COLUMN_ID	COLUMN_NAME	FORMATTED_DATATYPE	NULLABLE	DATA_DEFAULT
1	1	INCIDENT_ID	NUMBER	N	"DBI5"."ISEQ$$_76372".nextval
2	2	ORI_CODE	VARCHAR2(7)	N	<null>
3	3	INCIDENT_YEAR	NUMBER(4,0)	N	<null>
4	4	INCIDENT_MONTH	NUMBER(2,0)	N	<null>
5	5	INCIDENT_NUMBER	VARCHAR2(20)	N	<null>
6	6	SITUATION	VARCHAR2(60)	N	'Single Victim/Single Offender'
7	7	HOMICIDE_TYPE	VARCHAR2(60)	N	'Murder and Nonnegligent Manslaughter'
8	8	VICTIM_COUNT	NUMBER(3,0)	N	1
9	9	OFFENDER_COUNT	NUMBER(3,0)	N	1
10	10	INCIDENT_DATE	DATE	N	<null>

Constraints:

	CONSTRAINT_NAME	CONSTRAINT_TYPE	COLUMN_NAME	SEARCH_CONDITION	REFERENCED_TABLE
1	CHK_INCIDENT_HOM_TYPE	CHECK	HOMICIDE_TYPE	homicide_type IN ('Murder and Nonnegligent Mans…	<null>
2	CHK_INCIDENT_MONTH	CHECK	INCIDENT_MONTH	incident_month BETWEEN 1 AND 12	<null>
3	CHK_INCIDENT_OFF_CNT	CHECK	OFFENDER_COUNT	offender_count >= 0	<null>
4	CHK_INCIDENT_SITUATION	CHECK	SITUATION	situation IN ('Single Victim/Single Offender',…	<null>
5	CHK_INCIDENT_VIC_CNT	CHECK	VICTIM_COUNT	victim_count >= 1	<null>
6	CHK_INCIDENT_YEAR	CHECK	INCIDENT_YEAR	incident_year BETWEEN 1900 AND 2100	<null>
7	SYS_C008538	CHECK	INCIDENT_ID	"INCIDENT_ID" IS NOT NULL	<null>
8	SYS_C008539	CHECK	ORI_CODE	"ORI_CODE" IS NOT NULL	<null>
9	SYS_C008540	CHECK	INCIDENT_YEAR	"INCIDENT_YEAR" IS NOT NULL	<null>
10	SYS_C008541	CHECK	INCIDENT_MONTH	"INCIDENT_MONTH" IS NOT NULL	<null>
11	SYS_C008542	CHECK	INCIDENT_NUMBER	"INCIDENT_NUMBER" IS NOT NULL	<null>
12	SYS_C008543	CHECK	SITUATION	"SITUATION" IS NOT NULL	<null>
13	SYS_C008544	CHECK	HOMICIDE_TYPE	"HOMICIDE_TYPE" IS NOT NULL	<null>
14	SYS_C008545	CHECK	VICTIM_COUNT	"VICTIM_COUNT" IS NOT NULL	<null>
15	SYS_C008546	CHECK	OFFENDER_COUNT	"OFFENDER_COUNT" IS NOT NULL	<null>
16	SYS_C008547	CHECK	INCIDENT_DATE	"INCIDENT_DATE" IS NOT NULL	<null>
17	PK_INCIDENT	PRIMARY KEY	INCIDENT_ID	<null>	<null>
18	FK_INCIDENT_AGENCY	FOREIGN KEY	ORI_CODE	<null>	AGENCY
19	UQ_INCIDENT_CASE	UNIQUE	ORI_CODE	<null>	<null>
20	UQ_INCIDENT_CASE	UNIQUE	INCIDENT_YEAR	<null>	<null>
21	UQ_INCIDENT_CASE	UNIQUE	INCIDENT_MONTH	<null>	<null>
22	UQ_INCIDENT_CASE	UNIQUE	INCIDENT_NUMBER	<null>	<null>

3.3 Berechtigungen von Rollen und Benutzern

Berechtigungen der Rolle DataAnalyst:

Tx: Manual DDL

	ROLE	TABLE_NAME	PRIVILEGE
1	DATAANALYST	HISTORICAL_COMPARISON_VIEW	SELECT

Rechte von Benutzer PublicViewer:

	GRANTEE	TABLE_NAME	PRIVILEGE	GRANTOR
1	PUBLICVIEWER	HISTORICAL_COMPARISON_VIEW	SELECT	DBI5

Rollenzuweisung an Benutzer Investigator:

	GRANTEE	GRANTED_ROLE	ADMIN_OPTION	DEFAULT_ROLE
1	INVESTIGATOR	DATAANALYST	NO	YES

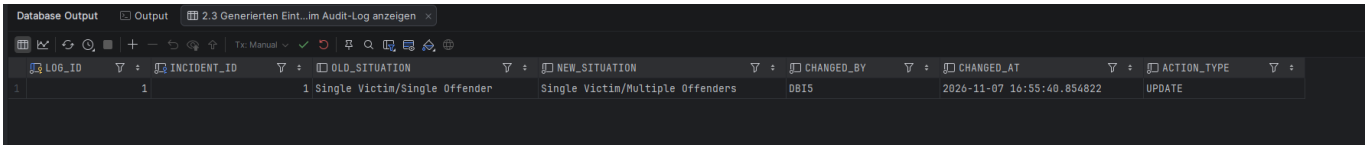
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4. Trigger (Kapitel 2.6)

4.1 Before-Insert-Trigger: Altersvalidierung (0–120 Jahre)

```
DBI5> INSERT INTO person (age, sex, race, ethnicity)
  VALUES (25, 'M', 'White', 'Not of Hispanic Origin')
1 row affected in 11 ms
DBI5> INSERT INTO person (age, sex, race, ethnicity)
  VALUES (150, 'M', 'Asian or Pacific Islander', 'Unknown')
Error : 2290, Position : 0, SQL = INSERT INTO person (age, sex, race, ethnicity)
VALUES (150, 'M', 'Asian or Pacific Islander', 'Unknown'), Original SQL = INSERT INTO person (age, sex, race, ethnicity)
VALUES (150, 'M', 'Asian or Pacific Islander', 'Unknown'), Error Message = ORA-02290: CHECK-Constraint (DBI5.CHK_PERSON_AGE) verletzt
DBI5> INSERT INTO agency (ori_code, agency_name, state_name, population, county)
  VALUES ('TEST001', 'Test Police Dept', 'Alaska', 50000, 'Test County')
1 row affected in 5 ms
DBI5> INSERT INTO incident (ori_code, incident_year, incident_month, incident_number, situation, homicide_type, victim_count, offender_count, incident_date)
  VALUES ('TEST001', 2025, 11, 'CASE-999', 'Single Victim/Unknown Offender(s)', 'Murder and Nonnegligent Manslaughter', 1, 0, DATE '2025-11-01')
1 row affected in 6 ms
DBI5> UPDATE incident
  SET situation = 'Single Victim/Single Offender',
      offender_count = 1
  WHERE ori_code = 'TEST001' AND incident_number = 'CASE-999'
1 row affected in 13 ms
```

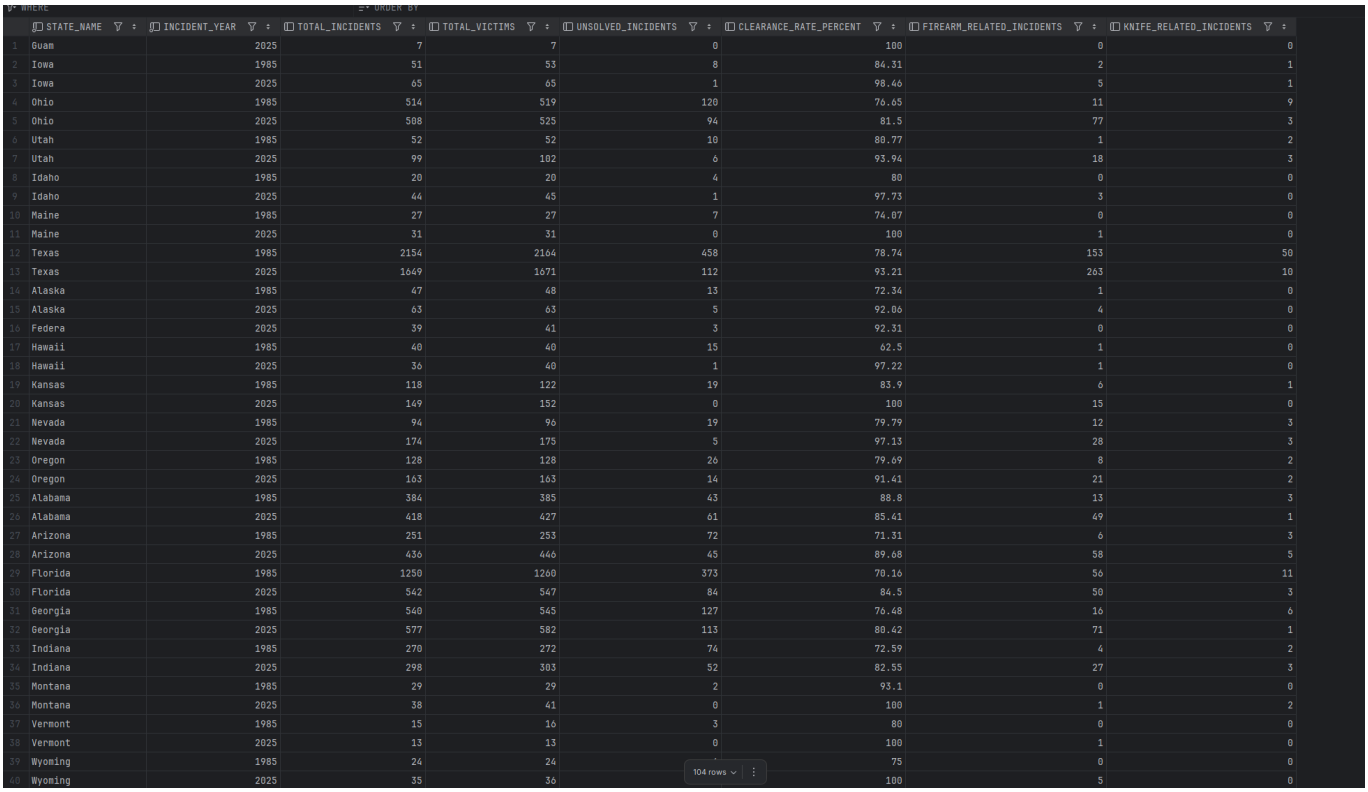
4.2 After-Update-Trigger: Audit-Log bei Statusänderung



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5. View (Kapitel 2.7)

Historical_Comparison_View



WHERE ORDER BY

	STATE_NAME	INCIDENT_YEAR	TOTAL_INCIDENTS	TOTAL_VICTIMS	UNSOLVED_INCIDENTS	CLEARANCE_RATE_PERCENT	FIREARM_RELATED_INCIDENTS	KNIFE_RELATED_INCIDENTS
1	Guam	2025	7	7	0	100	0	0
2	Iowa	1985	51	53	8	84.31	2	1
3	Iowa	2025	65	65	1	98.46	5	1
4	Ohio	1985	514	519	120	76.65	11	9
5	Ohio	2025	508	525	94	81.5	77	3
6	Utah	1985	52	52	10	80.77	1	2
7	Utah	2025	99	102	6	93.94	18	3
8	Idaho	1985	20	20	4	80	0	0
9	Idaho	2025	44	45	1	97.73	3	0
10	Maine	1985	27	27	7	74.07	0	0
11	Maine	2025	31	31	0	100	1	0
12	Texas	1985	2154	2164	458	78.74	153	50
13	Texas	2025	1649	1671	112	93.21	263	10
14	Alaska	1985	47	48	13	72.34	1	0
15	Alaska	2025	63	63	5	92.06	4	0
16	Federa	2025	39	41	3	92.31	0	0
17	Hawaii	1985	40	40	15	62.5	1	0
18	Hawaii	2025	36	40	1	97.22	1	0
19	Kansas	1985	118	122	19	83.9	6	1
20	Kansas	2025	149	152	0	100	15	0
21	Nevada	1985	94	96	19	79.79	12	3
22	Nevada	2025	174	175	5	97.13	28	3
23	Oregon	1985	128	128	26	79.69	8	2
24	Oregon	2025	163	163	14	91.41	21	2
25	Alabama	1985	384	385	43	88.8	13	3
26	Alabama	2025	418	427	61	85.41	49	1
27	Arizona	1985	251	253	72	71.31	6	3
28	Arizona	2025	436	446	45	89.68	58	5
29	Florida	1985	1250	1260	373	70.16	56	11
30	Florida	2025	542	547	84	84.5	50	3
31	Georgia	1985	540	545	127	76.48	16	6
32	Georgia	2025	577	582	113	80.42	71	1
33	Indiana	1985	270	272	74	72.59	4	2
34	Indiana	2025	298	303	52	82.55	27	3
35	Montana	1985	29	29	2	93.1	0	0
36	Montana	2025	38	41	0	100	1	2
37	Vermont	1985	15	16	3	80	0	0
38	Vermont	2025	13	13	0	100	1	0
39	Wyoming	1985	24	24		75	0	0
40	Wyoming	2025	35	36		100	5	0

104 rows